

**Detecting Network Intrusions With an Autoencoder: Unsupervised Anomaly Detection on
NSL-KDD**

Ruthvik Nath Bandari

Northeastern University

AAI6680: AI for Cybersecurity

Mimoza Dimodugno, PhD

September 29, 2026

Detecting Network Intrusions With an Autoencoder: Unsupervised Anomaly Detection on NSL-KDD

Signature-based intrusion detection only catches attacks someone has already described. This lab tested an alternative: an autoencoder trained on normal network connections only, which flags any connection it reconstructs poorly. The official NSL-KDD test set includes 17 attack types absent from training, which tests generalization to unseen attacks.

Method

NSL-KDD (Canadian Institute for Cybersecurity, n.d.; Tavallaee et al., 2009) provides 125,973 training and 22,544 test connections, each described by 41 features. Following the brief’s numerical-only rule, I dropped the three categorical fields and the constant *num_outbound_cmds*, leaving 37 inputs rather than 41. The 67,343 normal training records were split 80/20 into training and validation sets. Heavy-tailed count and byte features were log-transformed, and a StandardScaler was fitted on the training portion only.

The autoencoder (37–20–10–20–37, ReLU hidden layers, linear output) was trained with Adam and mean squared error for up to 50 epochs with early stopping (patience 5), which never triggered. The anomaly threshold was the 95th percentile of training reconstruction errors. An Isolation Forest (Liu et al., 2008) received identical inputs and the same threshold rule.

Results

Training loss fell from 0.786 to 0.134, and validation loss followed it with no widening gap (Figure 1). The threshold flagged 5.02% of unseen normal validation records. On the test set, the autoencoder achieved precision 0.957, recall 0.730, F1 0.828, and ROC-AUC 0.958, with a 4.3% false-positive rate (Figure 2).

The autoencoder beat the Isolation Forest on ROC-AUC and F1 (Table 1). The Isolation Forest had higher precision and fewer false positives, and at a 1% false-positive budget it caught 56% of attacks against 48%. The autoencoder detected far more remote-to-local (R2L) and somewhat more user-to-root (U2R) attacks, while the Isolation Forest was better on scans. On

test-only attack types it detected 66.5% of attack records, versus 75.7% on seen types (Isolation Forest: 69.0%).

Attack Type Analysis

Floods and scans such as *neptune*, *nmap*, and *satan* exceeded 90% recall (Figure 3). R2L was hardest, at 32%, though category mappings vary between authors. *snmpgetattack* and *snmpguess* were never detected; at least 30% of *snmpgetattack* records are identical to normal records on all 37 features, so these features cannot separate them.

Two misses point to how the error is scored. *guess_passwd* reached only 11%. Most of its records show no failed login, but 38% do, a value found in only 0.1% of normal training records, and at least 70% of those were still missed. Similarly, 88% of *pod* records (27% recall) have a non-zero wrong-fragment count, never seen in normal training traffic. In both cases, averaging the error over 37 features probably diluted a strong single-feature signal. *smurf* (2%) connections carry a median of 1,008 source bytes, against 30 for normal ICMP echo-reply connections, but without the protocol and service fields that volume probably looks ordinary. An ablation showed that the log transform raised ROC-AUC from 0.937 to 0.958.

Reflection

In Module 1, my Gaussian mixture detector missed SSH brute force and bot traffic because each individual flow looked normal. I expected the autoencoder to fail on *guess_passwd* for the same reason, and partly it did: most password guesses look like ordinary single sessions. But the evidence showed a second cause. Many guesses carried a rare failed-login signal, and the model still missed most of them, probably because averaging the error across 37 features hid it. The lesson I take from this lab is that detection depends on how anomalies are scored, not only on how flexible the model is.

The lab also reinforced that the threshold is a security decision, not a technical detail. Moving from the 95th to the 99th percentile cuts false alarms from 4.3% to under 1%, but drops recall from 73% to 47%. That trade-off belongs to the team answering the alerts. The work I want to do is build detectors that model behavior over time, per user and per host, and pair them

with analysts who can act on ranked, explained alerts. An autoencoder is a useful part of that system, but not the whole system.

References

- Canadian Institute for Cybersecurity. (n.d.). *NSL-KDD dataset* [Data set]. University of New Brunswick. <https://www.unb.ca/cic/datasets/nsl.html>
- Liu, F. T., Ting, K. M., & Zhou, Z.-H. (2008). Isolation forest. In *Proceedings of the 2008 Eighth IEEE International Conference on Data Mining* (pp. 413–422). IEEE.
<https://doi.org/10.1109/ICDM.2008.17>
- Tavallaei, M., Bagheri, E., Lu, W., & Ghorbani, A. A. (2009). A detailed analysis of the KDD CUP 99 data set. In *Proceedings of the 2009 IEEE Symposium on Computational Intelligence for Security and Defense Applications* (pp. 1–6). IEEE.
<https://doi.org/10.1109/CISDA.2009.5356528>

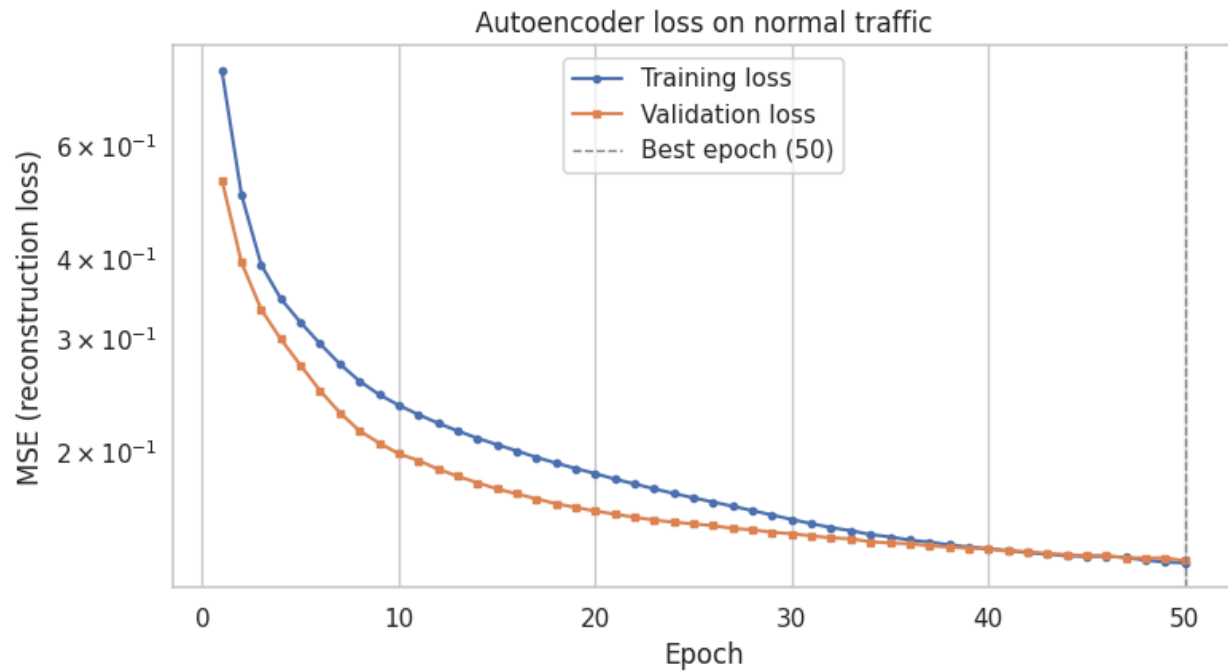
Table 1*Autoencoder Compared With Isolation Forest on the NSL-KDD Test Set*

Metric	Autoencoder	Isolation Forest
Precision	0.957	0.972
Recall	0.730	0.660
F1-score	0.828	0.786
False-positive rate	0.043	0.025
ROC-AUC	0.958	0.939
Recall at 1% false-positive rate	0.481	0.561
Recall: DoS / Probe	0.855 / 0.815	0.801 / 0.914
Recall: R2L / U2R	0.318 / 0.695	0.067 / 0.500

Note. Both models were trained on the same normal-only records and thresholded at the 95th percentile of their own training scores. The better value in each row is in bold.

Figure 1

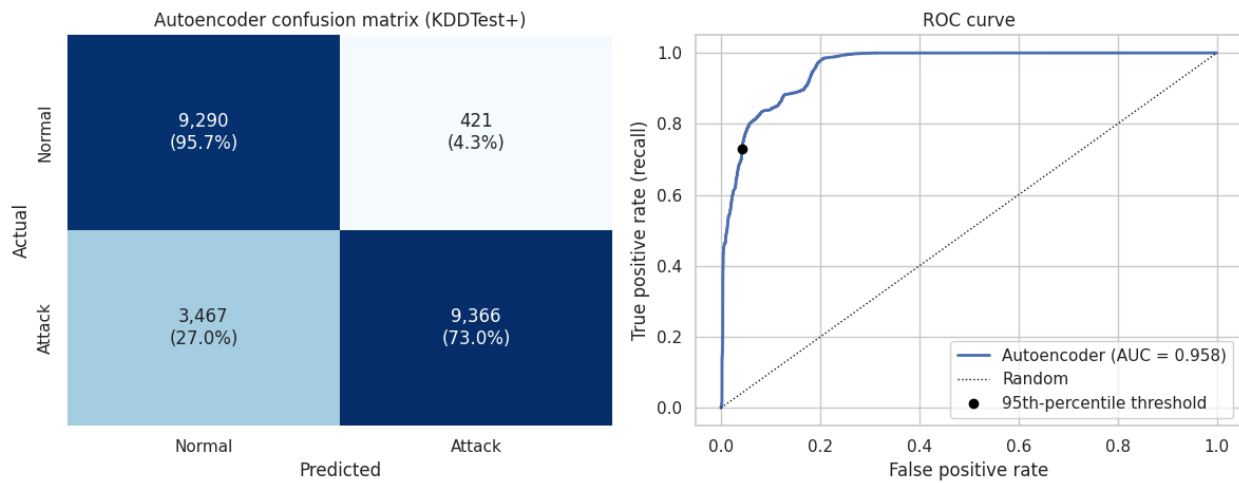
Autoencoder Training and Validation Loss on Normal Traffic



Note. Mean squared reconstruction error per epoch on a logarithmic scale. Both sets contain normal records only; validation loss was still improving at epoch 50, so early stopping did not end training.

Figure 2

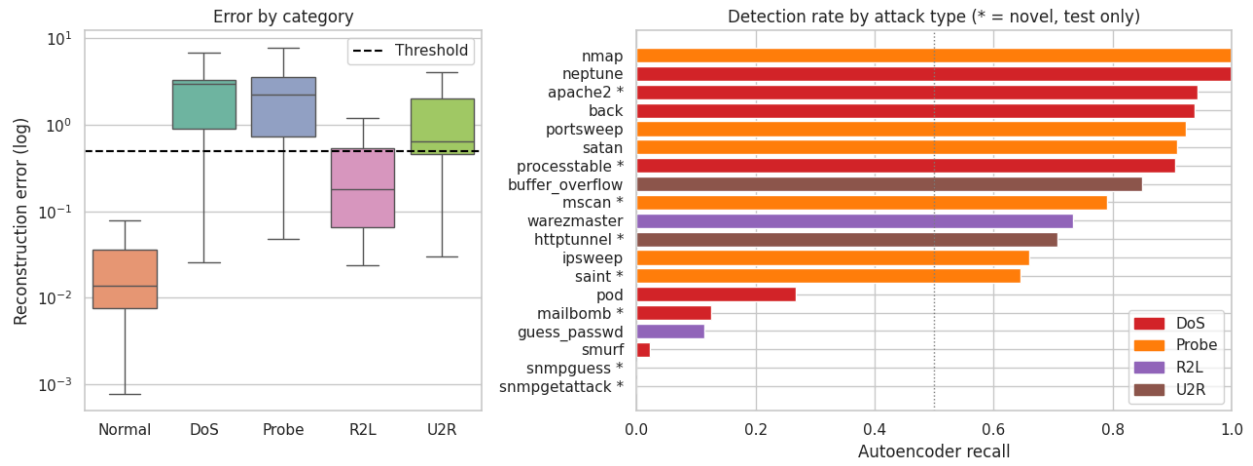
Autoencoder Confusion Matrix and ROC Curve on the NSL-KDD Test Set



Note. Threshold = 95th percentile of training reconstruction error. The black dot marks this operating point on the ROC curve. Test set: 9,711 normal and 12,833 attack connections.

Figure 3

Reconstruction Error by Category and Detection Rate by Attack Type



Note. Left: error distributions on a log scale; the dashed line is the detection threshold. Right: autoencoder recall for attack types with at least 20 test records; an asterisk marks types absent from training.